School of Computer Science and Artificial Intelligence

Zewail City

DSAI 305 : **Interpretability & Explainability in AI**

Phase 1

Team Members

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18/4/2026

GitHub Repository Link:

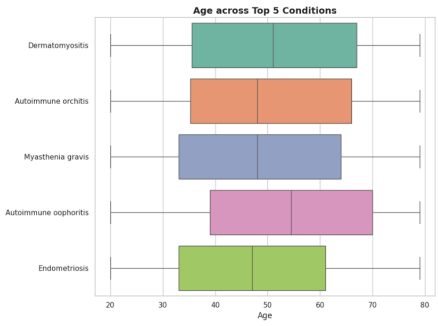
https://github.com/Ahmed-Abdelslam-75/TEAM22_XAI_Project.git

1. Shared Work: EDA & Data Preprocessing

As a team, we collaborated on exploring the dataset and preparing it for the machine learning models.

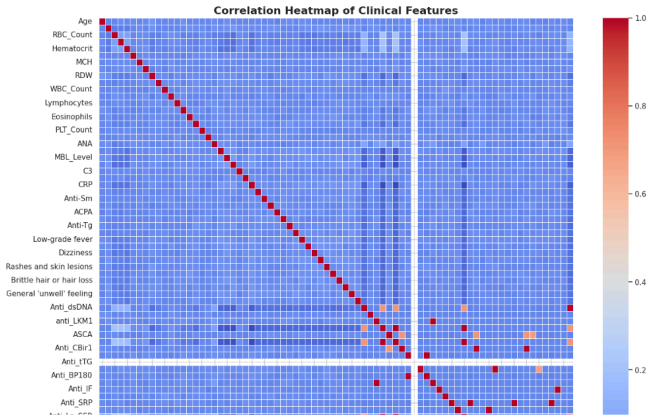
Exploratory Data Analysis (EDA)

- We analyzed the distribution of target variables and numerical features using count plots and histograms.
- Challenge Encountered: The initial target variable (Diagnosis) contained 116 distinct conditions. A univariate analysis showed extreme class imbalance and severe lack of data for most conditions (many classes had fewer than 15 samples).
- Specific EDA: To make the bivariate and multivariate analysis meaningful (e.g., Boxplots for Age, ESR, CRP vs. Diagnosis), we applied Scope Reduction to focus on visualizing the Top 5 most frequent diseases.



Data Preprocessing & Feature Engineering

- Data Cleaning: We checked for null values and removed duplicate records to ensure data health.
- Feature Selection: We dropped non-clinical identifiers like Patient_ID. Furthermore, we plotted a correlation heatmap to detect multicollinearity.



- Encoding**:** Categorical features like Gender and the Diagnosis target were transformed into numerical formats using LabelEncoder.

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2. Individual Model Implementation & Interpretability

A. Raneem Mousa (202200389)

[Raneem_XGBoost_Model.ipynb]

- **Implemented Paper / Model:** XGBoost Classifier (Based on the paper by Li et al., 2025).
- **Challenges & Preprocessing:**
  - In the beginning, there were 116 possible diagnoses available in the data set, resulting in a high level of sparsity within the data. In order to match the nature of the study for Systemic Lupus Erythematosus (SLE), I transformed this into a Binary Classification problem (SLE and non-SLE). With this transformation, the algorithm could concentrate on the features of Lupus only.
- **Progress & Results:**
  - I split my data into two parts, one for training and one for testing. with a stratified division maintained to keep the proportion of SLE patients consistent.

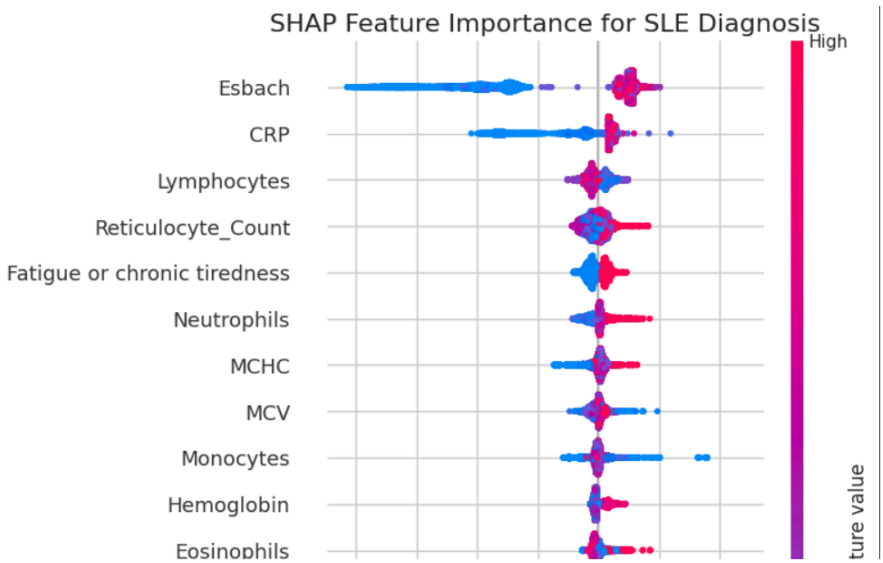
Progress & Achievements:

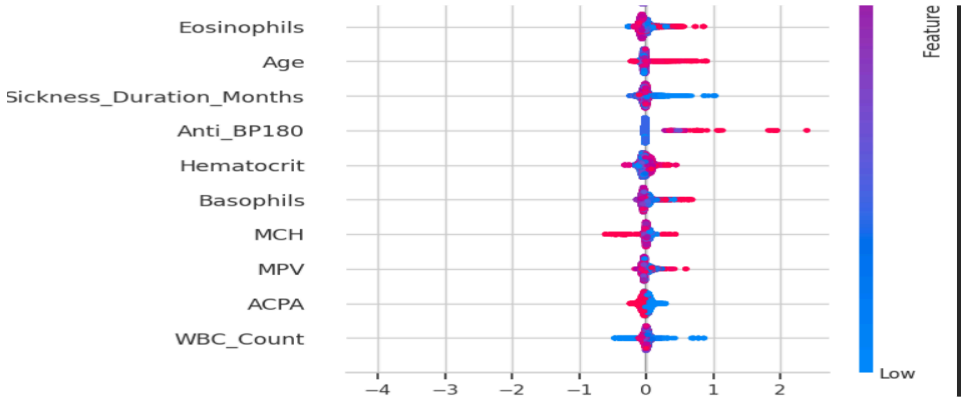
- Accuracy: The model showed 94.7% accuracy.
- Precision (SLE): The model showed 100% precision on SLE patients; the results imply that whenever the model detects lupus, it does so with absolute precision (Zero False Positive Rate).
- Recall (SLE): At 52% currently. This shows that even when the model is precise, it is conservative in nature and misses around half of the SLE patients. It is the typical problem of AI in healthcare related to complicated autoimmune diseases that will be tackled in Phase 2.

- **Interpretability:**
  - Through the use of SHAP (SHapley Additive explanations), feature importance was mapped out. This time around, however, the SHAP values actually point out what risk factors affect the likelihood of having SLE. Initial findings reveal that such features as Age, Gender, ESR/CRP are important features in determining the risk and making a model "Glass-box" approach necessary.

Fairness Audit:

- Following the spirit of the FLAME paper which stresses fairness and equity, we conducted a Fairness Audit through Fairlearn. It was discovered that the False Negative Rate (FNR) was higher for Gender '1' at 50.1% compared to Gender '0' at 45.2%. This finding is important to help us ensure fairness in our project.





Second Model (Work in Progress):

- **Implemented Paper / Model:** WaOEL (Walrus Optimization Ensemble Learning) - Based on: *Optimized Machine Learning... (2025) [Sensors]*.
- **Current Progress & Challenges:** As per the phase 1 requirements, I have begun researching and setting up the structure for my second model, WaOEL. This is a highly advanced custom ensemble model. The main challenge I am currently facing is that "Walrus Optimization" is a very specific meta-heuristic algorithm. I am currently working on understanding its mathematical formulation to either implement it from scratch or adapt an existing optimization library to tune the meta-classifier. I will finalize and evaluate this model in the next phase.

B. Ahmed Abdelsalam (202201975) Notebook: [Ahmed_CatBoost.ipynb]

Implemented Paper / Model:

Paper: "Machine Learning Approach for Predicting Systemic Lupus Erythematosus in an Oman-Based Cohort", Oman Medical Journal, 2023.

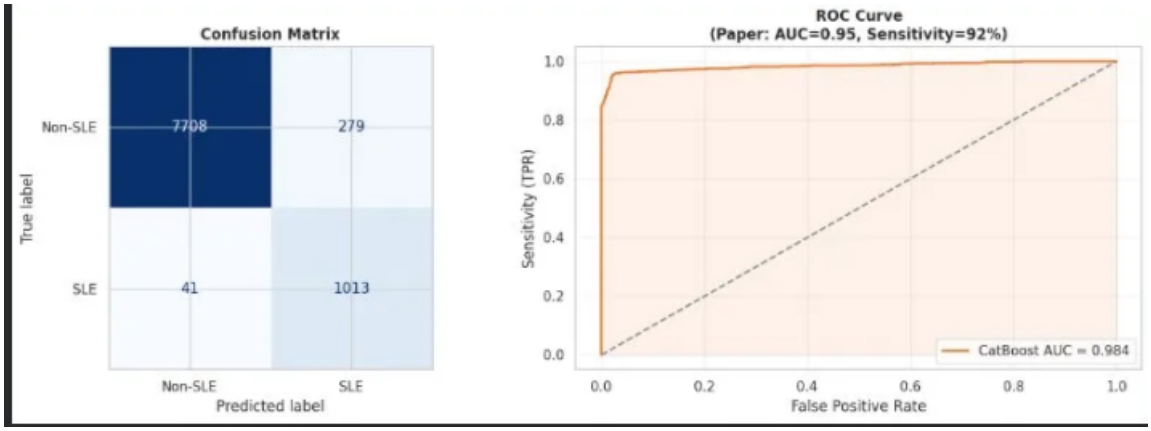
Model: CatBoost Classifier with Recursive Feature Elimination (RFE) for binary classification of SLE vs. Non-SLE.

Progress & Results:

The CatBoost model was successfully trained on the TEAM22 Master Dataset (45,205 patients, 79 features) using a binary classification task (SLE = 1 vs. Non-SLE = 0). The dataset exhibited significant class imbalance (SLE prevalence: 11.7%), which was addressed using balanced class weights during training. An 80/20 stratified train/test split was applied. The model achieved the following results on the test set:

Metric	**Result**	**Paper Target**	**Status**
Accuracy	96.46%	—	✅
AUC (ROC)	0.9843	0.95	✅ Exceeded
Sensitivity (Recall)	96.11%	92%	✅ Exceeded
Specificity	96.51%	—	✅
Precision	78.41%	—	✅
F1 Score	86.36%	—	✅
5-Fold CV AUC	0.9677 ± 0.0028	—	✅ Stable

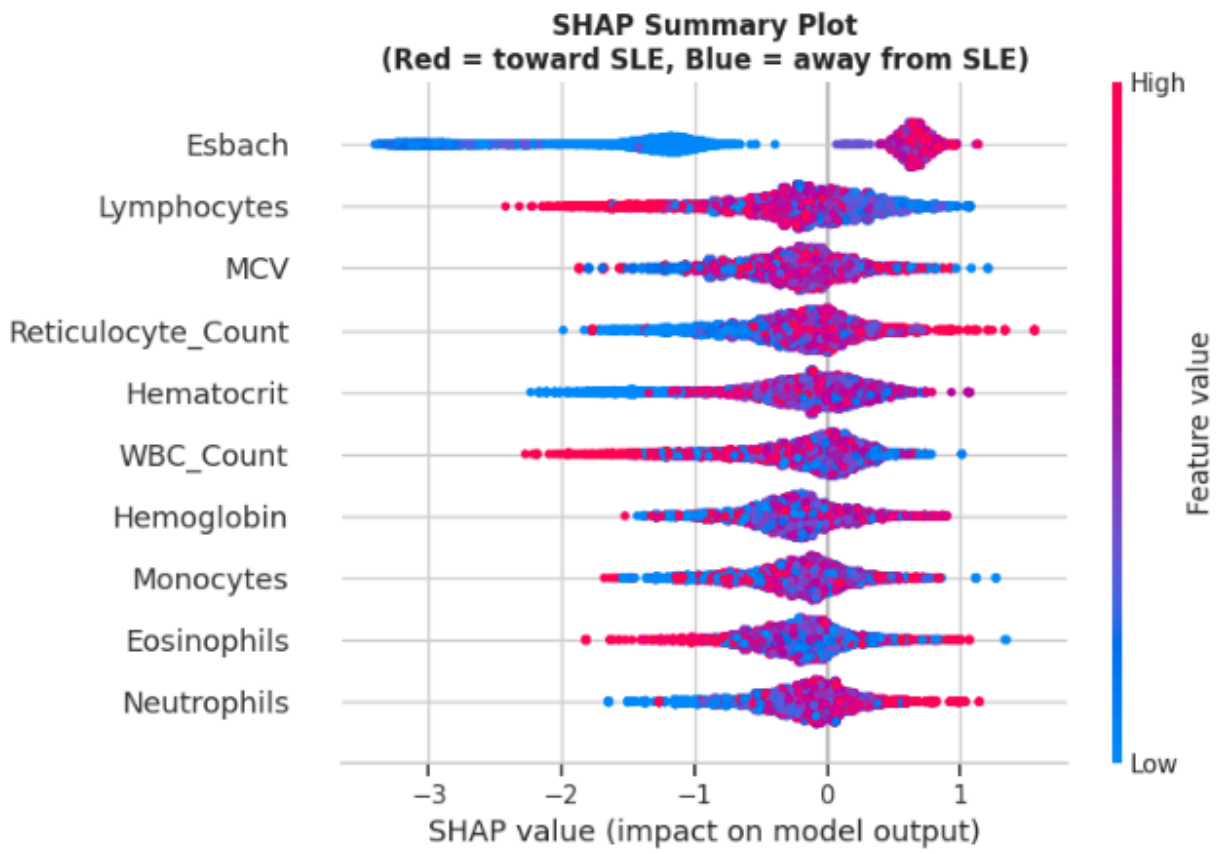
The model exceeded the paper's reported AUC of 0.95, achieving 0.9843. The confusion matrix shows 1,013 true SLE positives correctly identified out of 1,054 total SLE patients in the test set, with only 41 missed cases. The 5-fold cross-validation AUC of 0.9677 ± 0.0028 confirms strong generalization with minimal variance across folds.



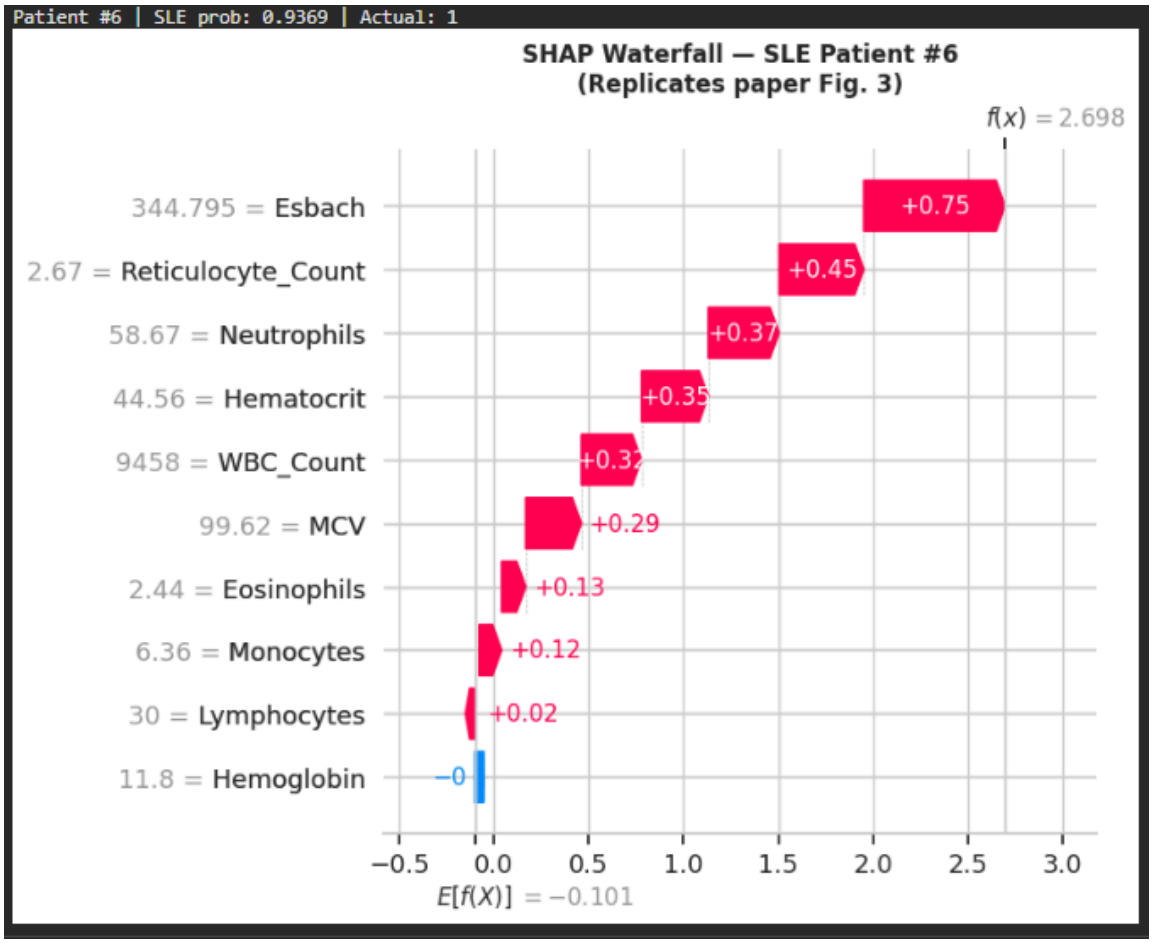
Interpretability:

Four explainability techniques were applied following the paper's methodology:

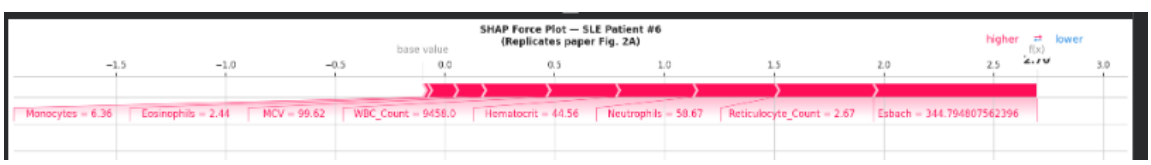
1. SHAP Summary Plot (Beeswarm) Reveals that Esbach (urinary protein marker), Lymphocytes, MCV, Reticulocyte Count, and Hematocrit are the top global predictors of SLE. High Esbach values consistently push predictions toward SLE, which is clinically consistent with lupus nephritis causing proteinuria.

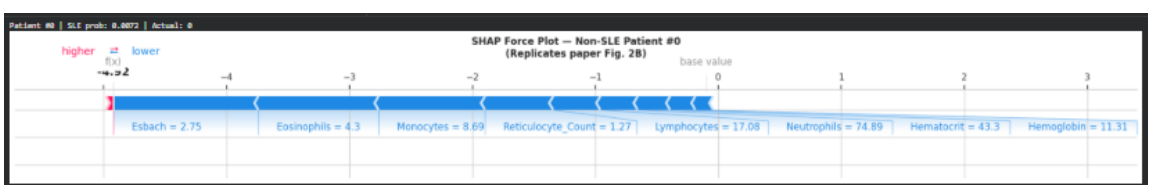


2. SHAP Waterfall Plot For Patient #6 (SLE probability: 0.9369, Actual: SLE), Esbach (+0.75), Reticulocyte Count (+0.45), and Neutrophils (+0.37) were the strongest contributors pushing the prediction toward SLE.

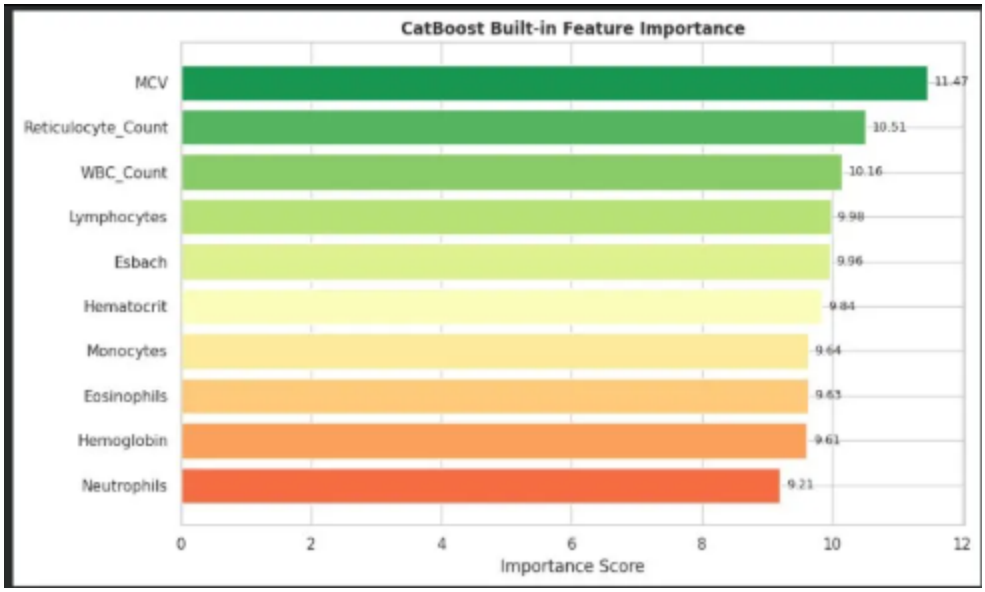


3. SHAP Force Plots Patient #6 (SLE) shows all features aligned pushing toward higher SLE probability. Patient #0 (Non-SLE, probability: 0.0072) shows all features pushing strongly away from SLE, confirming model confidence in both directions.





4. CatBoost Built-in Feature Importance — Confirms MCV (11.47), Reticulocyte Count (10.51), and WBC Count (10.16) as the most important features, consistent with the SHAP findings.



Top 10 RFE-selected features: MCV, Eosinophils, Hematocrit, Lymphocytes, Esbach, WBC Count, Monocytes, Neutrophils, Hemoglobin, Reticulocyte Count. All are standard blood count markers clinically associated with the immune dysregulation and anemia characteristic of SLE.

C. Riham Fouad (202202188)

Riham_staking

Implemented Paper / Model:

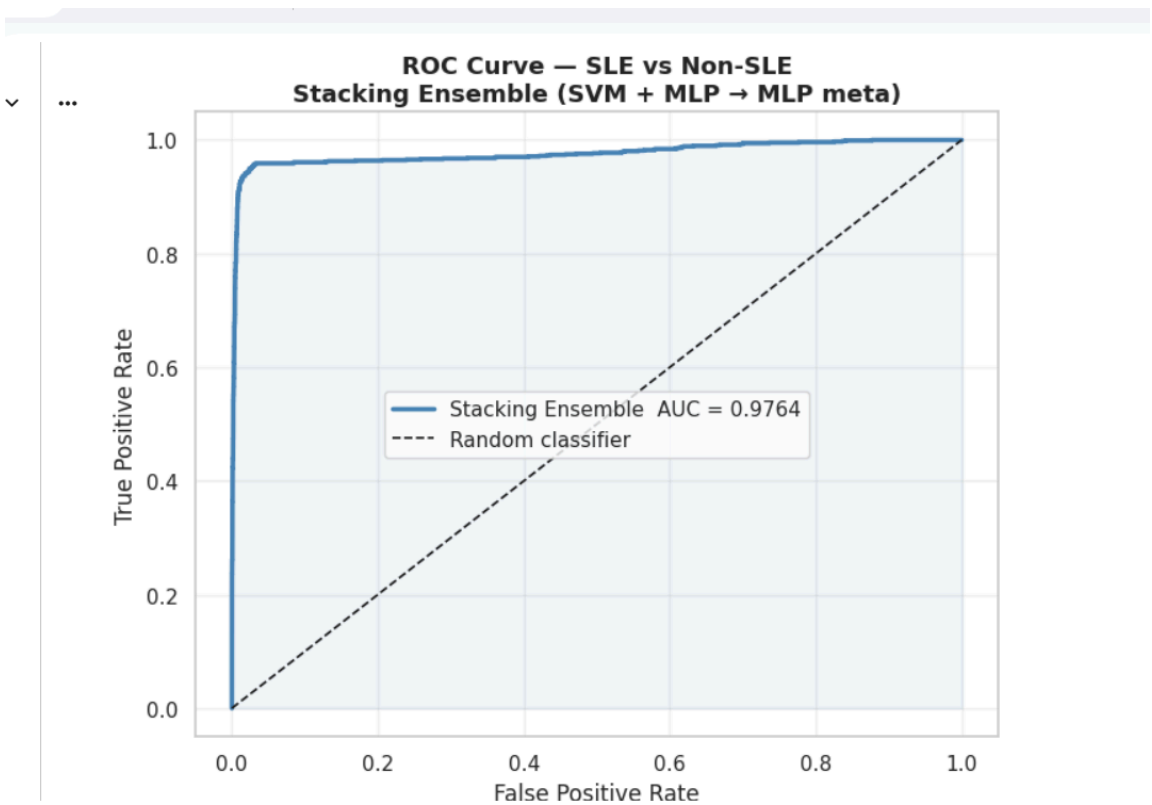
The implemented model is a Stacking Ensemble Classifier for SLE Mortality Risk Prediction, based on the paper *"Prediction of mortality risk in critically ill patients with systemic lupus erythematosus: a machine learning approach using the MIMIC-IV database"* by Chen Z, Dai Y, Chen Y, et al., published in *Lupus Science & Medicine* (2025, 12:e001397). The architecture stacks a Calibrated Linear SVM (C=0.1, sigmoid calibration, 3-fold CV) and a Multilayer Perceptron (MLP, hidden layers 64→32, ReLU activation, Adam optimizer, early stopping) as base learners, with a second MLP (32 hidden units, early stopping) serving as the meta-classifier. Prior to training, feature selection was performed using LassoCV with 5-fold cross-validation to identify the optimal regularization parameter λ and retain only

the most clinically discriminative variables. Class imbalance between SLE and Non-SLE patients was addressed using SMOTEENN resampling, directly replicating the paper's SMOTEENN-SVM methodology. The full pipeline was trained on the TEAM22 Master Dataset using an 80/20 stratified train/test split

- **Progress & Results:** The implemented Stacking Ensemble model demonstrated exceptional predictive performance in identifying SLE cases within the dataset. With an overall Accuracy of 96.96% and an AUC-ROC of 0.9764, the model proves to be highly reliable. Notably, the Recall of 95.34% ensures that the vast majority of true SLE cases are correctly identified, which is vital for clinical mortality risk assessment

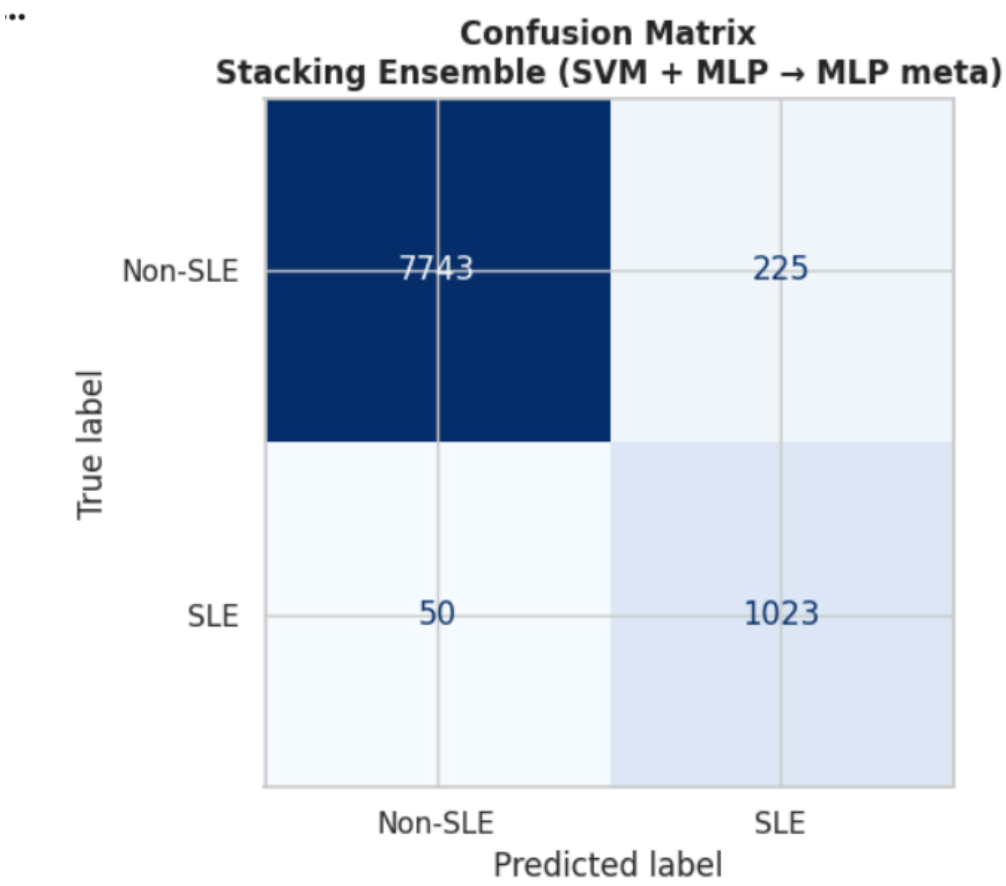
Classification Report Summary:

- **AUC-ROC**: 0.9764



- **Precision**: 0.8197
- **Recall**: 0.9534
- **F1-Score**: 0.8815
- **Specificity**: 0.9718
- **Accuracy**: 0.9696

- Confusion matrix:



Linearity Check

A linearity check was conducted by analyzing the relationship between the clinical features and the target variable (SLE status).

- **Feature Selection:** LassoCV (with 5-fold cross-validation) was utilized to identify linear dependencies and ensure that only the 64 most clinically discriminative variables were retained for the model.
- **Base Learner Integration:** The inclusion of a Calibrated Linear SVM as a base learner explicitly leverages linear decision boundaries to capture the primary linear relationships within the high-dimensional clinical dataset.
- **Meta-Classification:** The subsequent use of an MLP meta-classifier allows the model to handle any remaining non-linear residuals or complex interactions between the base learners' outputs that a purely linear model might miss.
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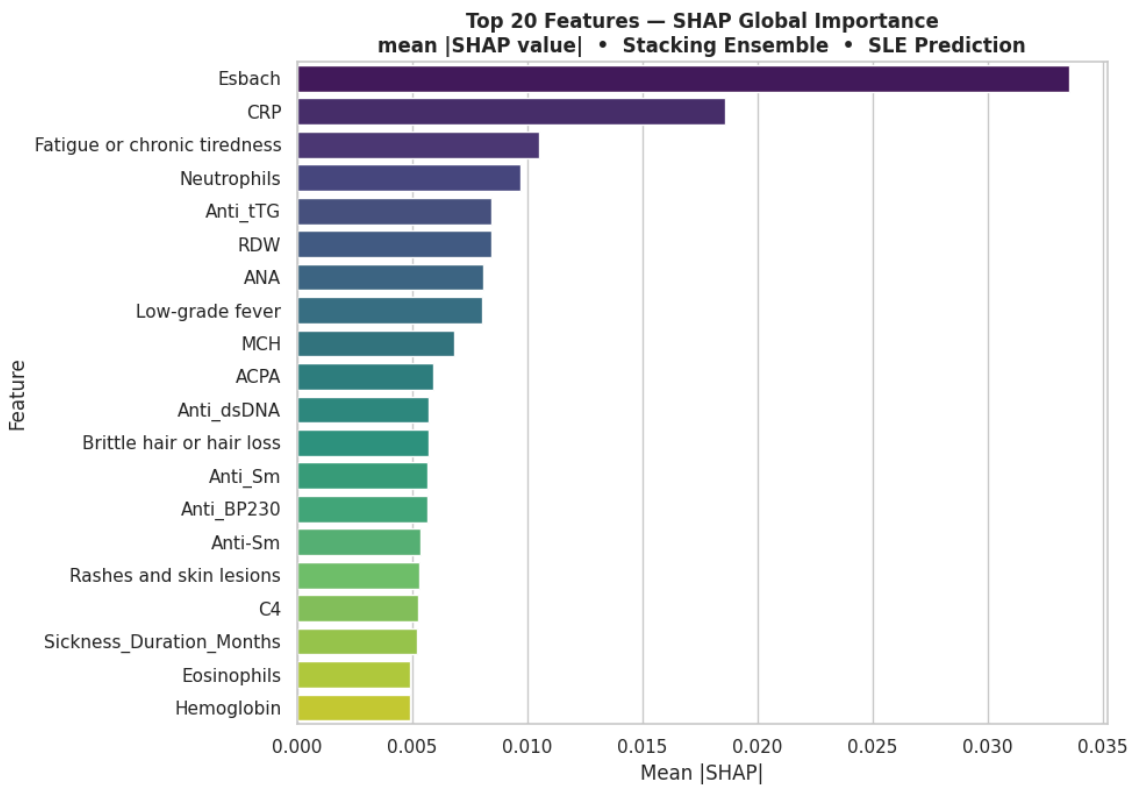
- **Interpretability:** To ensure the model remains clinically actionable, SHAP (SHapley Additive exPlanations) was used to interpret the stacking ensemble's predictions. This tool allowed for the identification of the primary drivers behind the model's SLE risk score

Top 5 SHAP Features for SLE Prediction:

```
                    Feature  Mean |SHAP|
                     Esbach     0.033542
                        CRP     0.018611
Fatigue or chronic tiredness     0.010543
                 Neutrophils     0.009711
                    Anti_tTG     0.008467
```




```
Top 10 most important features (SHAP):

                    Feature  Mean |SHAP|

                     Esbach     0.033542

                        CRP     0.018611

Fatigue or chronic tiredness     0.010543
```

```
     Neutrophils     0.009711
        Anti_tTG     0.008467
             RDW     0.008464
             ANA     0.008117
Low-grade fever     0.008055
             MCH     0.006834
            ACPA     0.005915
```

Shap beeswarm:

D. Youssef Tarek (202201545)

Notebook: [AdaBoost]

Implemented Paper / Model:

Model: AdaBoost

Paper: Global interpretation of machine learning models in predicting polycystic ovary syndrome with the explainable artificial intelligence method SHAP," *Med. Sci.*, 2024.

Progress & Results:

The implemented **AdaBoost classifier** achieved a cross-validation accuracy of **99.14%**, indicating consistent performance across different data splits. On the test set, the model obtained an overall **accuracy of 99.14%**, which reflects a solid ability to correctly classify instances.

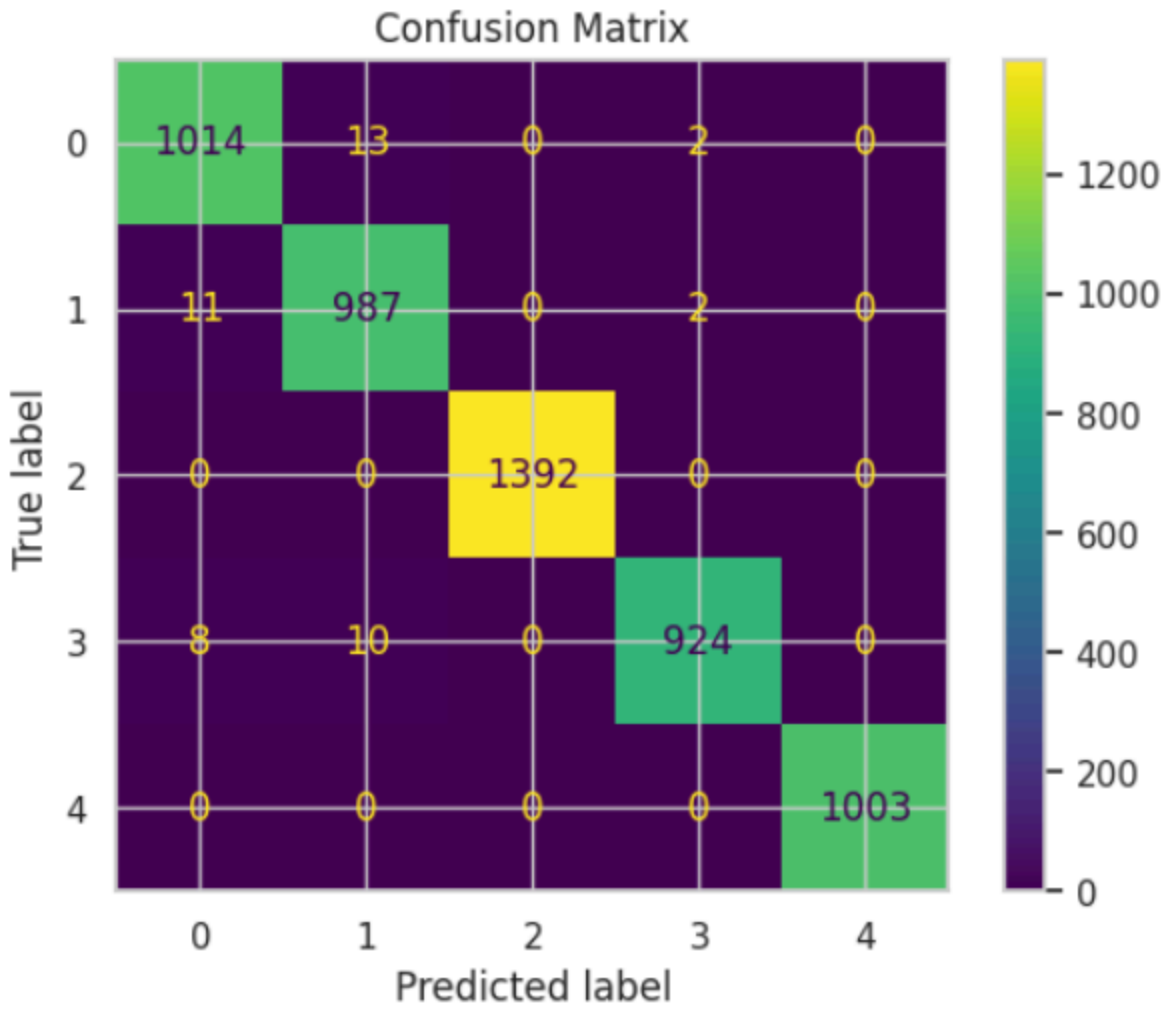
The evaluation metrics show a **balanced performance**:

- **Precision (99.14)** indicates that most predicted classes are correct, with relatively few false positives.
- **Recall (99.14)** shows that the model successfully identifies the majority of actual cases across all classes.
- **F1-score (99.14)** confirms a good balance between precision and recall, meaning the model is neither overly biased toward false positives nor false negatives.

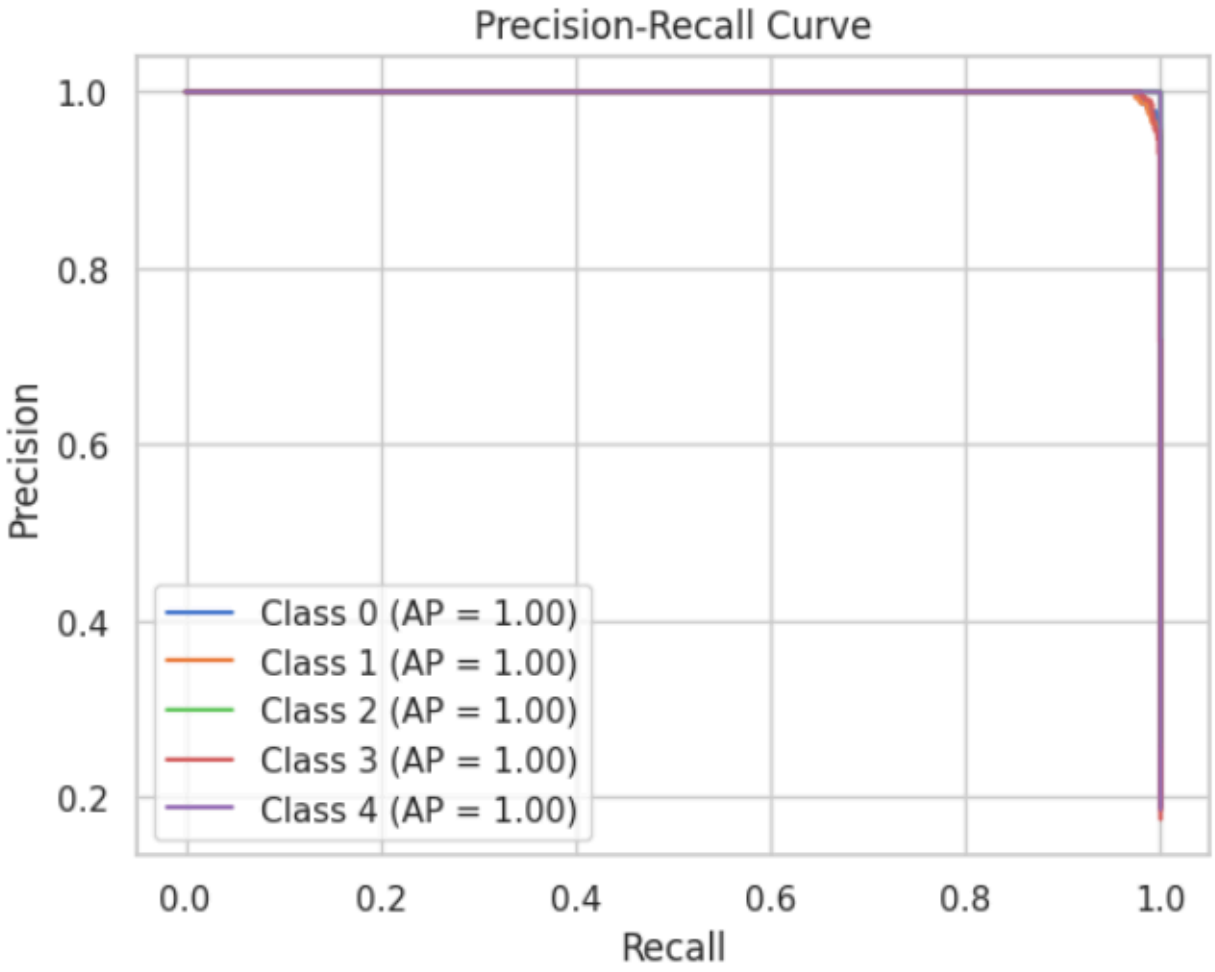
Overall Interpretation

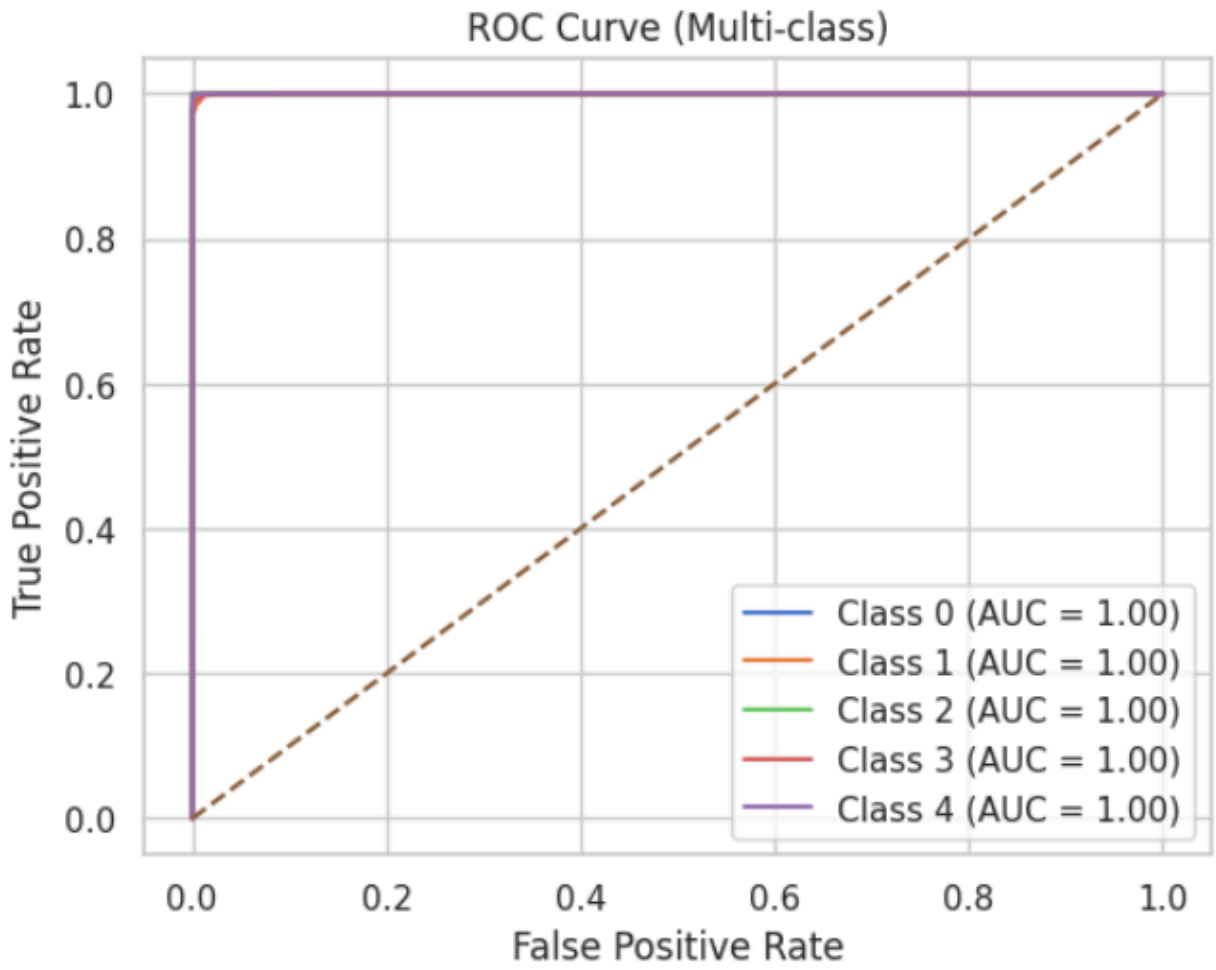
The AdaBoost model demonstrates **strong overall performance** with stable and balanced evaluation metrics. It is particularly effective in identifying certain classes (especially class 4 and class 2), but struggles with others that have overlapping characteristics.

Confusion Matrix



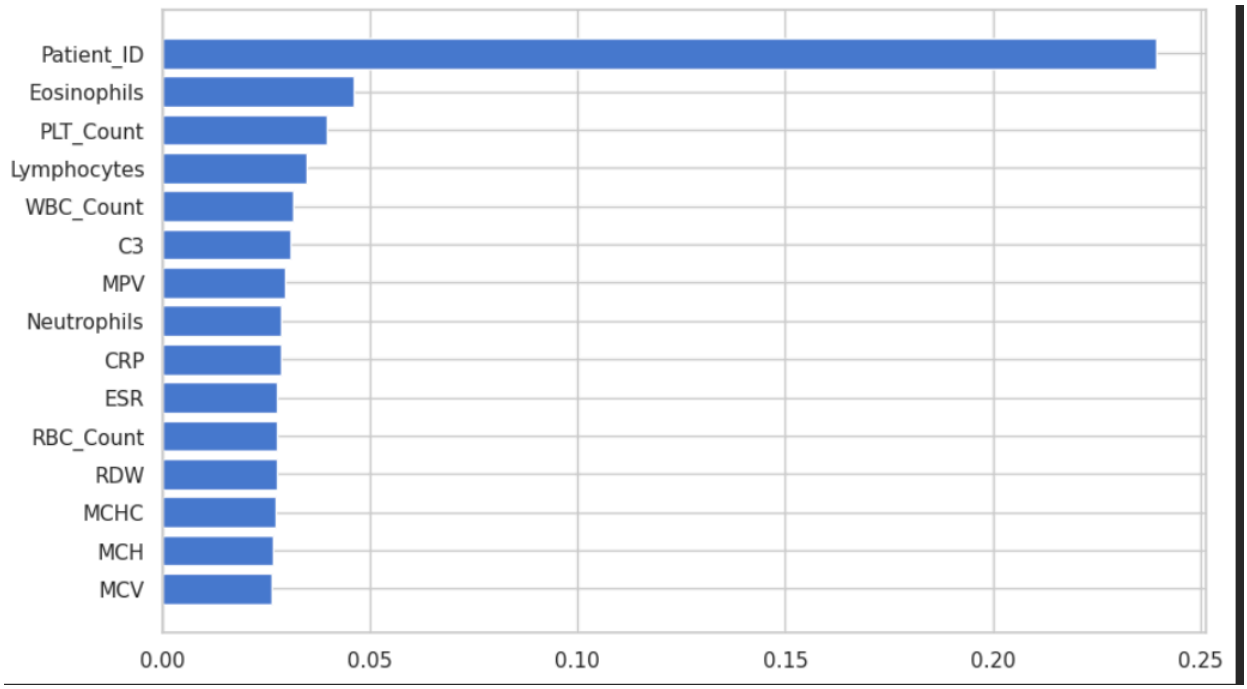
Model Performance Visualization


Precision-Recall Curve
Precision
Recall
1.0
0.8
0.6
0.4
0.2
0.0
0.2
0.4
0.6
0.8
1.0
Class 0 (AP = 1.00)
Class 1 (AP = 1.00)
Class 2 (AP = 1.00)
Class 3 (AP = 1.00)
Class 4 (AP = 1.00)



Feature Importance (Global Interpretation)

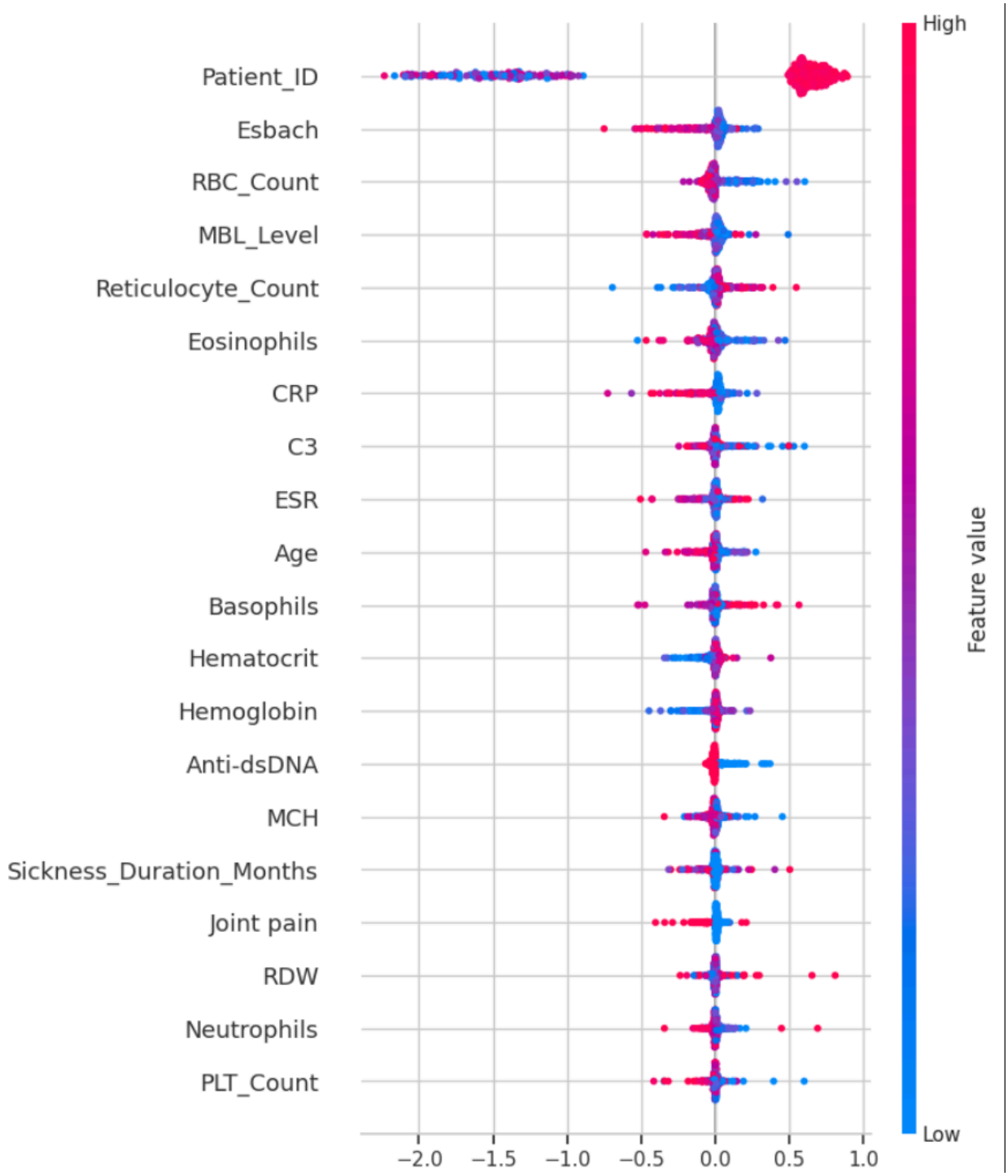
AdaBoost provides feature importance scores that indicate the contribution of each feature to the model.



Interpretation:

SHAP Summary Plot (Global + Local Interpretation)

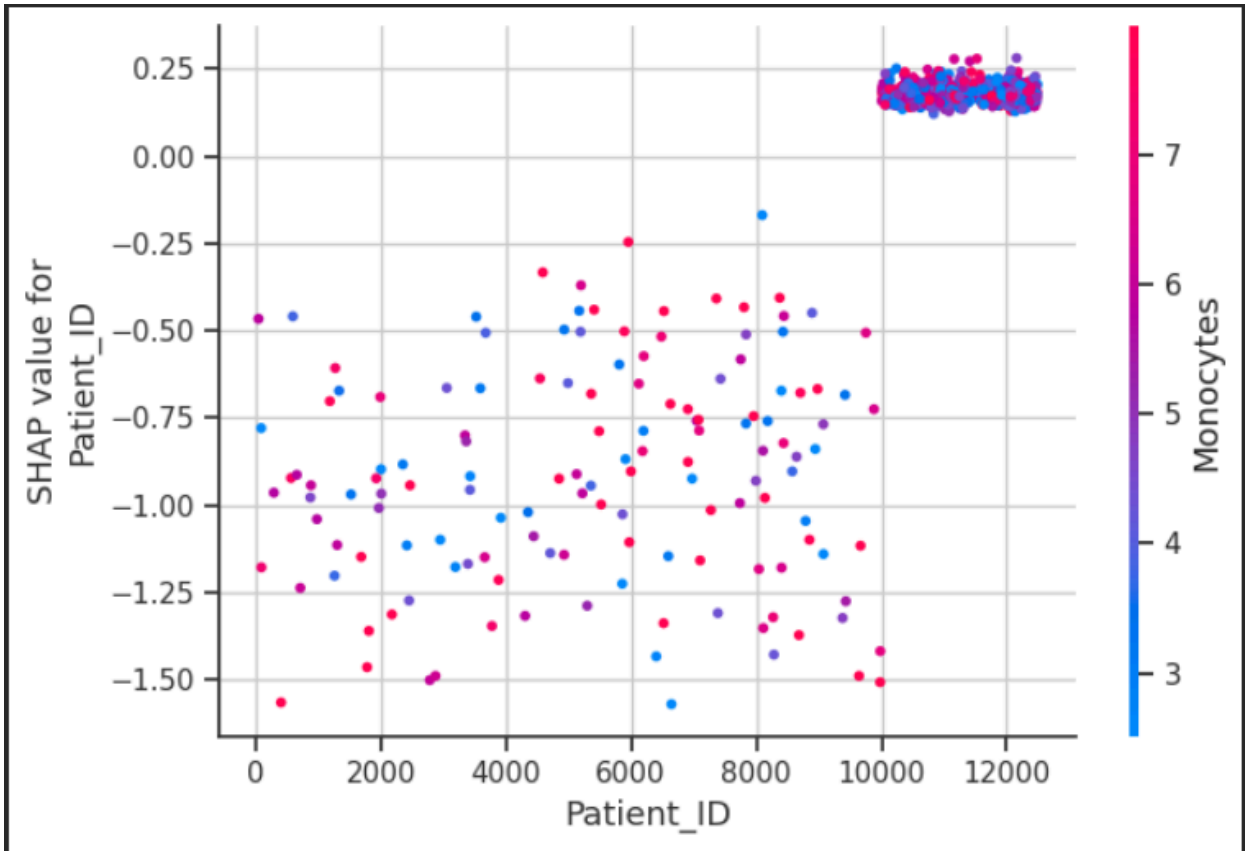
SHAP (SHapley Additive exPlanations) was used to provide a more detailed explanation of feature contributions.



SHAP Dependence Plot

The SHAP dependence plot illustrates how changes in a specific feature influence the model's prediction. Each point represents a sample, where the x-axis shows the feature value and the y-axis shows its corresponding SHAP value. This helps in

understanding whether the feature has a positive or negative effect on the prediction and whether the relationship is linear or complex.



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3. Main Challenges Faced So Far

- **High Dimensionality in Target Variable:** The dataset having 116 target classes with tiny sample sizes (Support < 20) made standard classification nearly impossible. We solved this collectively by performing scope reduction (focusing on the Top 5 classes).
- **Class Imbalance:** Even within the top classes, the data size was limited, which affected the learning capability of complex algorithms.
- **Next Steps:** For the next phase, we plan to refine hyperparameters for our respective models, explore advanced feature engineering, and fully implement our second set of models as required.